UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

No electronic/communication devices are permitted.

No exam materials may be removed from the exam room.

Electrical and Computer Engineering

ENEL301-18S2 (C) Fundamentals of Engineering Economics and Management

Examination Duration: 150 minutes

Exam Conditions:

Closed Book exam: Students may not bring in any written or printed materials.

Calculators with a ‘UC' sticker approved.

Materials Permitted in the Exam Venue:

None

Materials to be Supplied to Students:

1 x Standard 16-page UC answer booklet

Instructions to Students:

- **Attempt all Questions**
- Answer each question on new page
- Aim to provide clear, concise and complete answers, using examples were necessary. You do not need to write all answers in complete sentences: where appropriate the use of bullet points or tables will be acceptable.
- Work that is illegible will not be marked.
- Poor spelling and grammar may be penalised.
- Key equations and tables are provided at the end of this paper.

Question 1 [10 marks]

You plan to install an electric vehicle fast charging station in your front yard. Not only can you charge your own vehicle quickly, but you plan to allow other people in your neighbourhood to use it for a fee. At negligible cost, you have written an app that will let people book and pay online.

You are comparing two possible fast charge systems. The Edison Superfast is more expensive than the KwikCharge to buy and install, and will require more maintenance at frequent intervals in its early years of operation. However, it charges much more quickly and you predict that this option will be significantly more popular.

The resulting cash flows for the two options are shown below.

Your opportunity cost of capital is 10%.

	C_0, $	**C_1, $**	**C_2, $**	**C_3, $**	**C_4, $**
KwikCharge	-5,000	1,250	2,500	2,000	2,000
Edison Superfast	-20,000	5,000	2,000	8,000	16,000

a) What does the opportunity cost of capital represent?

b) Calculate payback for each project.

c) Calculate NPV for each project.

d) Which fast charger should you choose, and why?

Your neighbour, Wim, has just bought a Tesla Roadster, and wants to use your charging station. Wim doesn't like the idea of using your pay-as-you-go app, and proposes this deal instead. You will give Wim an annual credit for $100 of charger use a year for 10 years, and then Wim will pay you (or your beneficiaries) $100 a year forever.

e) Should you accept Wim's offer or not?

Question 2 **[10 marks]**

You are involved in estimating the costs of a new wind farm development for your employer, Hau Nui Energy. There are many different ways of making an estimate, but generally speaking they can be divided into two main categories.

a) Briefly describe the two main categories of cost estimates.

b) Discuss in general terms the expected accuracy of an estimate with respect to its purpose and where it occurs in the life cycle of a project.

As part of the overall estimate, you need to estimate the current cost of a piece of equipment that has 50% more capacity than a similar piece of equipment that cost $30,000 five years ago. The appropriate power-sizing exponent for this type of equipment is 0.8, and the ratio of the cost indexes (current to 5 years ago) is 1.24.

c) What is the cost of the equipment in today's dollars?

Hau Nui Energy is also involved in production of small scale domestic wind turbines.

The fixed cost of production of for the first 5000 units is $200,000. The variable costs at that level of production are $1800 per unit. Hau Nui plans to sell the turbines for $5000 each, directly to the consumer. Hau Nui senior management requires a target profit margin on total costs of 40% per unit. This is reflected in the selling price.

d) Calculate the breakeven number of units that must be sold in order to meet management expectations.

Question 3 [10 marks]

a) What are the three key elements that need to be proved in order to establish an act was legally negligent? Give a brief explanation of each.

b) There are six elements required for a contract to be valid. Name them, and explain what they mean.

c) As part of your business strategy you are considering patent protection for a new product. What are the essential features the product must have, and what steps must you take, in order for you to be able to secure that protection?

Question 4 [10 marks]

Fisher & Paykel Healthcare (FPH) is a NZ manufacturer, designer and marketer of products and systems for use in respiratory care, acute care, and the treatment of obstructive sleep apnea. Based in New Zealand, their products and systems are sold in around 120 countries worldwide.

Below are two extracts from FPH's 2018 Financial Statements. Use them to answer the questions below.

Fisher & Paykel Healthcare Corporation Limited

FINANCIAL STATEMENTS

CONSOLIDATED INCOME STATEMENT

For the year ended 31 March 2018

	Notes	2017 NZ$M	2018 NZ$M
Operating revenue	4	**894.4**	**980.8**
Cost of sales		(304.0)	(330.4)
Gross profit		590.4	650.4
Other income	5	5.0	5.0
Selling, general and administrative expenses		(269.3)	(290.9)
Research and development expenses		(86.0)	(94.7)
Total operating expenses		**(355.3)**	**(385.6)**
Operating profit before financing costs		**240.1**	**269.8**
Financing income		0.4	1.6
Financing expense		(3.5)	(3.7)
Exchange gain on foreign currency borrowings		1.5	0.1
Net financing expense		**(1.6)**	**(2.0)**
Profit before tax	5,11	**238.5**	**267.8**
Tax expense	11	(69.3)	(77.6)
Profit after tax		**169.2**	**190.2**

CONSOLIDATED BALANCE SHEET
As at 31 March 2018

	Notes	2017 NZ$M	2018 NZ$M
ASSETS			
Current assets			
Cash and cash equivalents		61.3	31.9
Short-term investments		–	100.4
Trade and other receivables	7	129.6	146.0
Inventories	8	135.0	125.4
Derivative financial instruments	6	21.2	18.8
Tax receivable		2.0	1.7
Total current assets		**349.1**	**424.2**
Non-current assets			
Derivative financial instruments	6	24.1	36.9
Other receivables		2.4	2.5
Property, plant and equipment	9	425.2	476.4
Intangible assets	10	44.5	50.4
Deferred tax assets	11	32.9	34.7
Total assets		**878.2**	**1,025.1**
LIABILITIES			
Current liabilities			
Interest-bearing liabilities	12	21.1	29.9
Trade and other payables	13	103.0	112.8
Provisions	14	3.2	4.7
Tax payable		14.7	22.0
Derivative financial instruments	6	3.6	9.0
Total current liabilities		**145.6**	**178.4**
Non-current liabilities			
Interest-bearing liabilities	12	39.9	52.5
Provisions	14	2.0	2.1
Other payables	13	8.6	8.6
Derivative financial instruments	6	5.1	4.9
Deferred tax liabilities	11	15.4	17.2
Total liabilities		**216.6**	**263.7**

	Notes	2017 NZ$M	2018 NZ$M
EQUITY			
Share capital	15	183.5	201.4
Treasury shares	15	(1.7)	(3.0)
Retained earnings		391.0	467.3
Reserves	17	88.8	95.7
Total equity		**661.6**	**761.4**
Total liabilities and equity		**878.2**	**1,025.1**

The accompanying Notes form an integral part of the Financial Statements.

On behalf of the Board
25 May 2018

Tony Carter
Chairman

Lewis Gradon
Managing Director and
Chief Executive Officer

a) What does EBIT stand for, and what does it tell us? What is Fisher and Paykel’s EBIT for the year ended 31 March 2018?

b) Calculate the current ratio for FPH for year ended 31 March 2018. What does this tell you about the financial operation of the company?

c) What is the accounting equation, and what does it mean? Calculate the debt/equity ratio for FHP.

d) Calculate the percentage ROE for Fisher and Paykel Healthcare for the years ended 31 March 2017 and 2018. Comment on your answer.

e) Define unsystematic (specific) risk. Name two unsystematic risk factors that the managers of Fisher and Paykel Healthcare should be assessing in their strategic planning. Include a brief explanation of your choice of factors.

Question 5 [10 marks]

Hau Nui Energy Ltd has identified land on the Summit Rd in the Christchurch Port Hills as suitable location for a wind farm development. The proposal is for 52 turbines, 2.3 MW each. Each tower is 68 m high, and total high of each turbine to top of blade is 117 m.

View of Christchurch from Summit Rd, Port Hills. PHOTO: GEOFF SLOAN

You are a senior engineer at Hau Nui, and you have been asked to lead a team to undertake a comprehensive triple bottom line analysis of the development.

a) What are the limitations of triple bottom line analysis in decision making for sustainable development?

As part of your analysis, you plan to perform an assessment of environmental effects (AEE). As a first step, identify:

b) the potential environmental effects of the wind farm.

c) the stakeholders affected by the proposal, and their potential concerns

Section 7 of the Resource Management Act 1991 states:

In achieving the purpose of this Act, all persons exercising functions and powers under it, in relation to managing the use, development, and protection of natural and physical resources, shall have particular regard to—

(a) Kaitiakitanga:

(aa) The ethic of stewardship:

(b) The efficient use and development of natural and physical resources:

(ba) the efficiency of the end use of energy:

(c) The maintenance and enhancement of amenity values:

(d) Intrinsic values of ecosystems:

(e) *Repealed*.

(f) Maintenance and enhancement of the quality of the environment:

(g) Any finite characteristics of natural and physical resources:

(h) The protection of the habitat of trout and salmon:

(i) the effects of climate change:

(j) the benefits to be derived from the use and development of renewable energy.

d) Briefly discuss the implications of Section 7 of the RMA for your AEE.

As part of your economic analysis, you need to calculate the cost of electricity generated by the wind farm.

You estimate that the total capital cost of the development is \$340 million, while operating and maintenance costs are \$15/MWh. The capacity factor is 40%, and the estimated life is 20 years. NZ Treasury is currently recommending a real discount rate of 7% for energy infrastructure projects.

e) Calculate the cost of generated electricity in cents per kWh.

Key Equations

$C_A/C_B = (S_A/S_B)^X$

$SP.x = VC(x) + FC + P$

$CM = TR – VC$

$PV_{perpetuity} = C/r$

$FV_t = PV(1+ i)^t$

$FV_t = PV(1 + i/m)^{mt}$

$EAR = (1 + i/m)^m – 1$

$1 + i_{real} = (1 + i_{nom})/(1 + infl)$

$Zu = Ku^n$ where $n = \log s / \log 2$

$PV_{growing\ perpetuity} = C/(r-g)$

$$PV_{growing\ annuity} = \frac{C}{r-g}\left[1 - \left(\frac{1+g}{1+r}\right)^t\right]$$

$$FV_{growing\ annuity} = C\left[\frac{(1+r)^t-(1+g)^t}{r-g}\right]$$

$$FVannuity = C\left[\frac{(1+i)^t - 1}{i}\right]$$

$$PVannuity = C\frac{1 - \frac{1}{(1+i)^t}}{i}$$